

NARAEN PALANIKUMAR

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SUMMARY

Full-Stack Bioinformatics AI Engineer and Technical Founder with an M.Eng. from UIUC and comprehensive expertise in computational biology, multi-omics, cloud-native data infrastructure, and AI architecture. Proven track record of developing scalable bioinformatics pipelines (scRNAseq, RNA-seq, spatial transcriptomics, proteomics, metabolomics), engineering complex ETL workflows, and orchestrating robust RAG architectures on Google Cloud. Adept at driving rapid prototyping and translating complex biological and unstructured datasets into peer-reviewed insights and end-to-end SaaS solutions.

EDUCATION

University of Illinois at Urbana-Champaign

Master of Engineering (M.Eng.) in Bioengineering

Urbana-Champaign, IL

May 2024

University of Illinois at Urbana-Champaign

Bachelor of Science (B.S.) in Bioengineering

Urbana-Champaign, IL

May 2023

EXPERIENCE

Founding Engineer @ Phalaena Automata

Remote

AI Infrastructure for Bioinformatics & Synthetic Biology

Full-Time, Mar. 2026 – Present

- Spearheading the end-to-end technical development of a data-intensive SaaS platform, managing the product lifecycle from initial architecture to cloud deployment.
- Architecting scalable bioinformatics workflows and deploying a multi-tenant cloud data platform on Google Cloud for diverse high-throughput Omics ingestion.
- Architecting full-stack, cloud-native AI infrastructure, integrating LLMs to automate unstructured data interpretation pipelines.

Independent Bioinformatics Consultant

Remote

Single-Cell & Metabolic Pathway Analysis

Per Diem, Jul. 2025 – Present

- Consulting on single-cell sequencing pipelines and metabolic pathway analysis to provide scalable workflows for research teams.
- Applying cloud-native bioinformatics pipelines built for high-throughput omics research.

Bioinformatics Research Assistant / Data Analyst

Bethesda, MD

Uniformed Services University of the Health Sciences (USUHS)

Full-Time, Oct. 2024 – Jun. 2025

- Led bioinformatic analysis of CD4 T-cell repertoires from patient sample data in the Malloy Lab, investigating respiratory infectious diseases.
- Managed exploratory data analysis (EDA) and optimized high-throughput processing of biological samples for flow cytometry and single-cell RNA sequencing.
- Developed custom software solutions to clean, normalize, and process vast tabular datasets, accelerating insights for high-impact journal publications.

Healthcare Technician

Champaign, IL

Carle Foundation Hospital, Emergency Department (Level I Trauma Center)

Part-Time, Sep. 2023 – Jul. 2024

- Managed patient care and diagnostic workflows using Epic EMR, performing critical tasks including EKGs, blood collection, and urinalysis.
- Facilitated seamless information flow between patients and providers, assisting across triage and specialized EKG technician roles.

Research Assistant (Computational Focus)

Champaign, IL

Sweeney Lab, University of Illinois at Urbana-Champaign

Per Diem, Aug. 2021 – Jun. 2023

- Analyzed complex scRNA-seq datasets using Cell Ranger and Seurat to uncover metabolism-associated neural circuits and rare cell populations.
- Built and launched an interactive web application from scratch to streamline large-scale photometry data processing and laboratory reproducibility.

Data Analyst & Research Assistant

Wheaton, IL

Thom Lab

Per Diem, May 2020 – Aug. 2020

- Led a comprehensive meta-analysis across 54 unique studies, designing R scripts for automated data extraction, transformation, and statistical validation.
- Designed and produced a 7-part training video series to onboard and upskill incoming lab members on software tools and analytical workflows.

PROJECTS

- Bench @ Phalaena Automata** | *React, FastAPI, Streamlit, Python, Nextflow, Docker, GCP* Mar. 2026 – Present
- Building a production-oriented platform for Omics analysis, supporting ingestion of various omics file formats into a centralized API.
 - Integrating Google Vertex AI via a tool-augmented RAG architecture, utilizing function calling to cross-reference processed data with PubMed APIs.
 - Containerizing complex genomic workflows using Nextflow and Docker, deployed on Google Cloud Run with CI/CD pipelines and strict IAM bucket permissions.

PUBLICATIONS

- Frontiers in Immunology:** "Patterns of restricted TCR usage following SARS-CoV-2 vaccination and severe disease" (Oct 2025).
- Bioinformatically isolated and modeled specific CD4 T-cells utilizing custom data extraction scripts to analyze TCR repertoires.
- The Journal of Neuroscience:** "Paraventricular Thalamic MC3R Circuits Link Energy Homeostasis with Anxiety-Related Behavior" (Sep 2023).
- Developed an interactive R Shiny application for fiber photometry data processing and visualization.

TECHNICAL SKILLS & CERTIFICATIONS

Bioinformatics: scRNA-seq, RNA-seq, Spatial Transcriptomics, Proteomics, Metabolomics, TCR Repertoires, 10x Genomics, Cell Ranger, Seurat, Nextflow, CWL

Programming: Python (FastAPI, Pandas, Scanpy, Biopython, SciPy), JavaScript/TypeScript (React, Next.js, Express), R, SQL, Shell, MATLAB

Cloud & Architecture: Google Cloud Platform (GCP), High-Performance Computing (HPC), Docker, UNIX/Linux, RESTful APIs

AI & Machine Learning: Vertex AI, LLMs, AI Architecture (RAG, Agents), PyTorch, Machine Learning (UMAP, KNN, Clustering)

Certifications & Lab: Google Cloud Professional Data Engineer, IBM Data Science Professional, Flow Cytometry, PCR, Biological Sample Processing

AWARDS & EXAM SCORES

Awards: National Merit Scholarship Finalist (2019), National AP Scholar (2019), USA Biology Olympiad Semifinalist (2018)

Scores (Optional): MCAT: 521 (98th %ile), ACT: 35 (99th %ile), SAT: 1550 (99th %ile)